



Capstone



#Lighting talk 1

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DSB-2



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Topic 1 CICIDS2017



Network traffic in SOC environments generates high volumes of alerts, many of which are false positives that waste analyst time. This project builds a machine learning classifier on the **CICIDS2017** dataset to distinguish benign traffic from real attacks and classify attack type, with the goal of reducing analyst triage time and false-positive fatigue a core problem in real-world SOC operations.



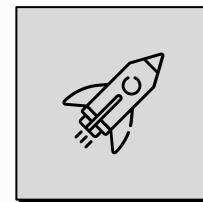


Topic 1: Goals & Approach



Goals

Cut SOC analyst triage time by separating benign traffic from real attacks, and classifying attack type



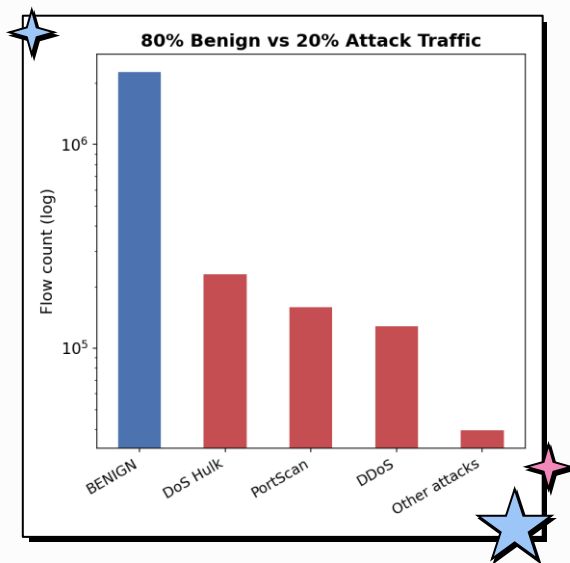
Approach

Multiclass classifier on CICIDS2017 flow data (unb.ca/cic), correcting for imbalance and redundant columns

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Key Finding



Traffic is severely imbalanced: BENIGN is 80% of flows, while rare attacks like Heartbleed have only 11 rows total

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Topic 1: EDA Highlights



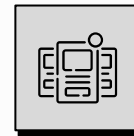
Target

Label: 15 classes, 80% BENIGN vs 20% attack traffic



Features

78 numeric features, plus Label and Source File



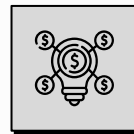
Nulls

Only Flow Bytes/s has NaNs 0.05% of rows affected



Correlation

15 duplicate columns found, e.g. Header Length.1



Outliers

8-23% of rows are outliers that IS the attack signal



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Topic 2 Mood Clustering

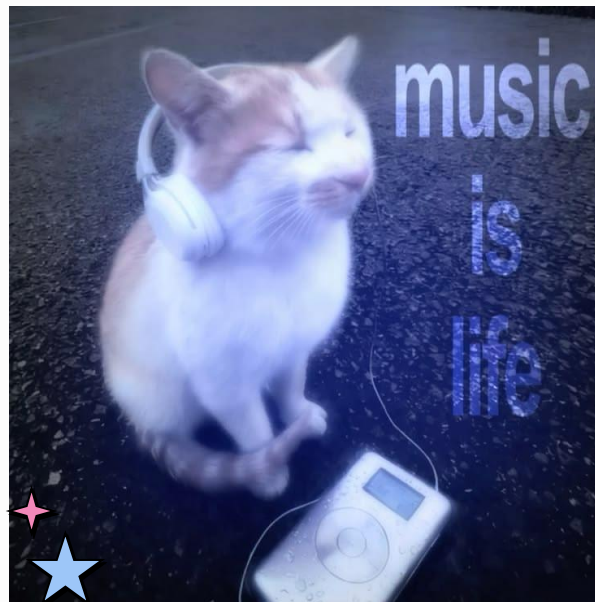


Without using genre labels, can songs be grouped into meaningful mood clusters from their audio DNA? This project applies unsupervised clustering (K-Means/DBSCAN) to Spotify audio features to discover natural song groupings, then interprets each cluster's dominant traits e.g., high energy + low valence = intense/dark as a foundation for a mood-based playlist generator.

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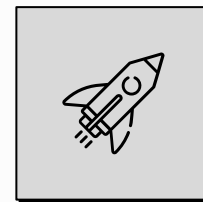


Topic 2: Goals & Approach



Goals

Discover natural mood groupings in songs without genre labels, as a base for a mood playlist generator



Approach

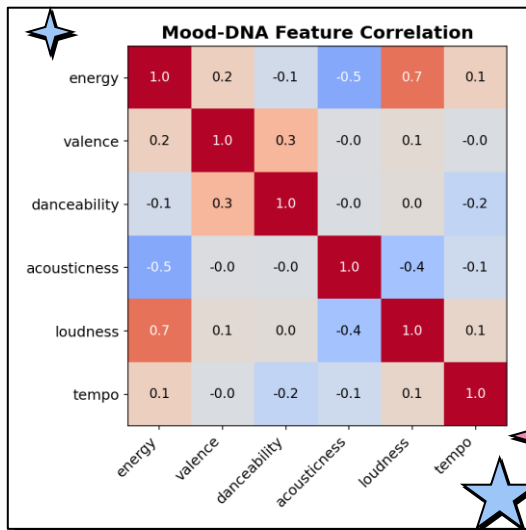
K-Means/DBSCAN on standardized audio DNA features (Kaggle spotify_songs), then interpret cluster traits

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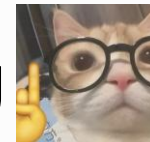


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Key Finding



Energy and loudness correlate at 0.68, the strongest redundancy among 7 candidate mood-DNA audio features

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Topic 2: EDA Highlights



Features

7 mood features: energy, valence, tempo, loudness+



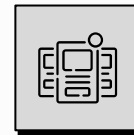
Scale

Loudness -46 to +1dB, tempo 0-239 needs scaling



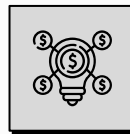
Weak Signal

Audio features barely predict popularity, corr 0.15



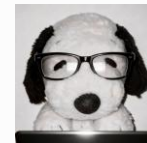
Redundancy

Energy and loudness correlate at 0.68, the top pair



Outlier

One track has tempo = 0, a failed-detection artifact



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Topic 3 CERT Insider Threat Detection Research

Insider threats employees misusing access to steal data, commit fraud, or sabotage systems are hard to detect since the activity looks normal in isolation. This project builds a **UEBA** pipeline that aggregates logon and device logs into per-user behavioral profiles, then flags employees whose behavior deviates from their baseline for SOC review.

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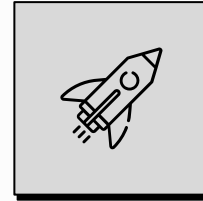


Topic 3: Goals & Approach



Goals

Flag employees whose behavior deviates from their own baseline, surfacing insider threats for SOC review



Approach

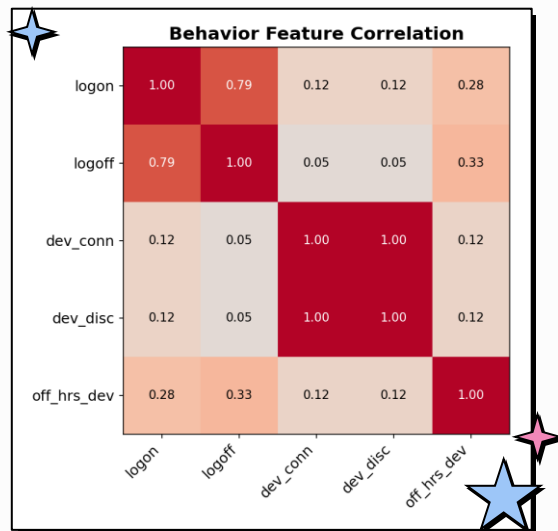
Aggregate logon/device logs (Kaggle CERT r4.2) into per-user profiles, then apply anomaly detection



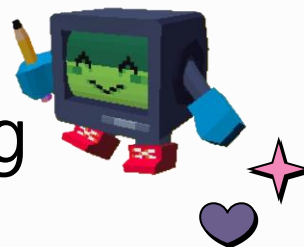
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Key Finding



device_connect and device_disconnect correlate at 0.999 a near-duplicate signal to simplify before modeling

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Topic 3: EDA Highlights



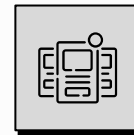
Scope

4,000 users, 1.39M user-days, Jan 2010-May 2011



Nulls

Zero nulls outer-join merge used fillna(0)



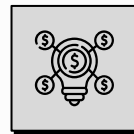
Redundancy

device_connect and disconnect correlate at 0.999



Sparsity

Every feature is zero-inflated most days show nothing



Anomaly

CRM0139: 19 off-hours device events in 1 day flagged



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Wrap-Up

All three datasets meet the 15k-row, 10-column bar

Where each topic stands:

- Topic 1: strongest signal, clear SOC use case
- Topic 2: most exploratory, no ground truth
- Topic 3: richest anomaly story to tell

Next: pick one topic and scope the full Capstone proposal



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Thanks!



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